

ALI BOZORGZADEH

CFD & Scientific Computing Enthusiast

 [abzrgzdh.github.io](https://github.com/abzrgzdh)
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EDUCATION

2018 – 2022

M.Sc. in Mechanical Engineering, BABOL NOSHIRVANI UNI. OF TECH.

- Thesis Topic: Investigation of the Effect of Flow Pulsation and Increased Discharged Voltage on Desalination and Energy Usage of Capacitive Deionization. ([source code](#))
- GPA: 3.65/4

2013 – 2018

B.Sc. in Mechanical Engineering, BABOL NOSHIRVANI UNI. OF TECH.

EXPERIENCE

2025 – present

Independent Open Source Development & Contribution

- [solids4foam/fluid-benchmarks](#): Contributed benchmark cases, verification studies, documentation and code development.
- [solids4foam/solids4foam](#), [OpenFOAM](#), [preCICE](#) (CalculiX adaptor): documentation, minor bug report and bugfix contribution

2022 – 2023

Research Assitant, BABOL NOSHIRVANI UNI. OF TECH.

- Particle separation using dielectrophoresis and a gradient in momentum due to a wall-mounted obstacle. ([paper](#), [source code](#))
- Updated legacy code to be compatible with newer versions of OpenFOAM.

2022

Teaching Assitant, BABOL NOSHIRVANI UNI. OF TECH.

- Prepared a series of [video tutorials](#) (in Persian) on developing custom OpenFOAM code.

PUBLICATIONS

2023

[1] Reza Derakhshan, **Ali Bozorgzadeh**, and Abas Ramiar. Numerical Investigation of Ternary Particle Separation in a Microchannel with a Wall-Mounted Obstacle Using Dielectrophoresis. *Journal of Chromatography A* (2023) DOI: [10.1016/j.chroma.2023.464079](https://doi.org/10.1016/j.chroma.2023.464079)

2020

[2] Masoud Outokesh, Seyed S. M. Ajarostaghi, **Ali Bozorgzadeh**, and Kurosh Sedighi. Numerical evaluation of the effect of utilizing twisted tape with curved profile as a turbulator on heat transfer enhancement in a pipe. *Journal of Thermal Analysis and Calorimetry* (2020) DOI: [10.1007/s10973-020-09336-0](https://doi.org/10.1007/s10973-020-09336-0)

COMPUTER SKILLS

Programming Languages

C, C++, Python, Bash, L^AT_EX, familiar with: x64 Assembly, Fortran, Typst, Lua, Elisp, JavaScript, Java

Libraries/Specifications

Numpy, Matplotlib, Pandas, Sympy, familiar with: MPI, OpenMP, OpenGL, OpenCL, Pthreads

CFD & CAD

OpenFOAM, Paraview, FreeCAD, Inkscape, Gnuplot familiar with: Ansys Fluent, Blender

Other tools

Linux, Git, Docker, SSH, Tmux

CERTIFICATES

ML Specialization, Cert.

- Supervised Machine Learning: Regression and Classification, Cert.
- Advanced Learning Algorithms, Cert.
- Unsupervised Learning, Recommenders, Reinforcement Learning, Cert.

LANGUAGES

English

- IELTS: 7.0/9.0 (R 8.0, R 7.5, W 6.0, S 7.0) (2023)
- TOEFL: 5.5/6.0 (R 5.5, L 5.5, W 5.5, S 5.5) (July 1, 2026)

Persian

Native language

REFERENCES

Abas Ramiar

Supervisor, co-author (associate professor)

Kurosh Sedighi

Course instructor, co-author (associate professor)

Amirhosein Ghasemi

Thesis advisor (assistant professor)